Volatility, Tail Risk, and Intraday Return Geometry: Evidence from Daily and High-Frequency Data

Johanna Rakotovao and Flavie Bertrand

December 20, 2025

1 Introduction

1.1 Motivation

Financial market risk manifests at different time horizons and through distinct statistical features. Major economic shocks may affect the average level of volatility, the persistence of volatility dynamics, the behavior of extreme returns, and the intraday process through which prices adjust to information. Capturing these dimensions requires a combination of complementary econometric tools rather than a single modeling framework. This paper examines market dynamics at daily and intraday frequencies, focusing on both typical fluctuations and extreme events.

(Topic 1) The first goal is to analyze the effect of the COVID-19 pandemic on the financial market in terms of volatility. This subject studies (i) pre- vs post-COVID volatility within each country and (ii) the US and China over the same period. The study period is from 2015 to 2024, with the start of the pandemic just in the middle. We chose to analyze China (CSI 300) and the United States (S&P 500), because they are two major economies with different structures in the world, that dealt with the crisis in different ways.
(Topic 2) Beyond average volatility dynamics, risk management requires quantifying *rare but severe* losses. Financial returns exhibit volatility clustering and heavy tails, so standard Gaussian-based risk measures may underestimate extreme events. In this section, we compare two complementary approaches to tail risk at the daily horizon: (i) a time-series model (AR(1)–GARCH(1,1)) producing time-varying volatility forecasts and conditional VaR under Gaussian innovations, and (ii) Extreme Value Theory (EVT) using a Peaks-Over-Threshold (POT) approach to estimate high-quantile VaR and ES from tail excesses.
(Topic 3) While daily returns and volatility summarize market behavior at a coarse time scale, high-frequency data provide a much richer description of intraday dynamics. In particular, the evolution of cumulative returns within a trading day can be viewed as a continuous curve, capturing how information is incorporated into prices over time. Rather than modeling individual high-frequency returns, a functional approach allows us to study the shape of intraday price movements and to identify common patterns across trading days. Functional Principal Component Analysis (FPCA) offers a natural framework to reduce the dimensionality of such curves while preserving their main modes of variation. This raises the question of whether intraday return curves exhibit systematic structures that can be summarized efficiently and whether these structures display any temporal dependence that could be exploited for forecasting.

1.2 Research questions

(Topic 1)

- Did the COVID-19 pandemic lead to a statistically significant change in the long-run average volatility of equity markets?
- Did the pandemic alter volatility persistence and the speed of mean reversion?
- Are the volatility responses to COVID-19 the same across countries ?

(Topic 2)

- Does a simple AR(1)–GARCH(1,1) model provide accurate short-term volatility forecasts when benchmarked against realised variance RV_t?
- For extreme risk (99% VaR/ES), does EVT-POT yield better calibration than a Gaussian GARCH-based VaR?

(Topic 3)

- Can intraday cumulative return curves be represented accurately in a low-dimensional functional space using FPCA?
- How should the number of functional principal components be chosen to balance model parsimony and goodness of fit?
- Do the resulting FPCA scores exhibit temporal dependence that allows meaningful forecasting of future intraday curves using simple time-series models?

1.3 Data overview

In this study, we use two datasets: (i) daily index data and (ii) intraday log-price data (`out28.csv`), which is used for both Topic 2 and Topic 3.
(Topic 1) The first dataset contains daily closing prices for the S&P 500 (United States) and the CSI 300 (China) indices. They are used to construct daily log returns over the period 2015–2024. The sample is split into pre-COVID (2015–2019) and post-COVID (2020–2024) subsamples. Returns are computed as log-differences of closing prices.
(Topic 2) The second dataset is intraday log-price data (`out28.csv`). From this intraday dataset, we construct two daily series: daily log returns R_t (close-to-close) and realised variance $RV_t = \sum_i r_{t,i}^2$ computed from intraday returns within day t. The sample contains $T = 753$ trading days. Realised variance serves as a nonparametric benchmark for daily volatility, while GARCH and EVT provide parametric approaches for volatility and tail risk.
(Topic 3) Using the same intraday dataset (`out28.csv`), we also study intraday dynamics. For each day, intraday log prices are observed at irregular time points and interpolated using the previous-tick method. From these data, we construct intraday cumulative log-return curves. Each intraday curve is evaluated on a regular grid of $J = 100$ time points over the trading day, allowing the data to be treated as functional observations.

2 Methodology

2.1 Volatility modelling (GARCH(1,1))

In financial data, volatility is not stable over time, and that is why we need to use a GARCH(1,1) model to compute the volatility. The volatility is time-varying.
We use a GARCH(1,1) model to estimate the conditional variance :

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim i.i.d.(0,1),$$
$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

Where :
- σ_t^2 is the conditional variance, so the volatility at time t give, the past
- α : impact of a return on the forthcoming volatility
- β : memory effect on the volatility
- Persistence $= \alpha + \beta$

2.1.1 Time-series model for daily returns (Topic 2)

We model daily log returns with a parsimonious AR(1)–GARCH(1,1) specification:

$$R_t - \mu = \phi(R_{t-1} - \mu) + \varepsilon_t, \qquad \varepsilon_t = \sigma_t Z_t, \qquad \sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2,$$

where (Z_t) are i.i.d. innovations with $\mathbb{E}[Z_t] = 0$ and $\mathbb{V}\eth\setminus(Z_t) = 1$ (Gaussian in the baseline implementation). The AR(1) component captures potential weak dependence in the conditional mean, while the GARCH(1,1) component captures volatility clustering through persistence $\alpha + \beta$.
We estimate parameters by (quasi-)maximum likelihood and assess fit using standard diagnostics: Ljung–Box tests on residuals and squared residuals and inspection of standardised residuals $z_t = \hat{\varepsilon}_t/\hat{\sigma}_t$.

2.2 Statistical inference : bootstrapping

Standard parametric tests are inappropriate for comparing average volatility across periods or countries, since the estimated conditional variance series $\hat{\sigma}_t^2$ is serially correlated by construction. To assess whether long-run volatility differs across groups, we test:
- $H_0 : E[\hat{\sigma}_{1,t}^2] = E[\hat{\sigma}_{2,t}^2]$,
- $H_1 : E[\hat{\sigma}_{1,t}^2] \neq E[\hat{\sigma}_{2,t}^2]$

where indices 1 and 2 show either different time periods (pre- vs post-COVID) or different countries over the same period.
To deal with serial dependence, inference is conducted using another method : block bootstrap. The estimated volatility series is divided into blocks of consecutive observations, which are resampled with replacement to generate bootstrap samples preserving the dependence structure. For each bootstrap replication $b = 1, \ldots, B$, the difference in mean volatility is computed as: $\Delta^{*(b)} = \bar{\hat{\sigma}}_1^{2*(b)} - \bar{\hat{\sigma}}_2^{2*(b)}$.
The empirical distribution of $\Delta^{*(b)}$ approximates the sampling distribution of the test statistic under dependence. The two-sided bootstrap p-value is given by:

$$p = \frac{1}{B}\sum_{b=1}^{B} \nVdash \left(|\Delta^{*(b)}| \geq |\Delta_{\text{obs}}|\right),$$

where $\Delta_{\text{obs}} = \bar{\sigma}_1^2 - \bar{\sigma}_2^2$.
If the p-value is small ($p < 0.05$), we reject H_0 and could conclude that the average volatility is significantly different between two periods.

2.3 Risk measures: VaR and ES

Conditional VaR under GARCH–Normal. Given the conditional distribution implied by the AR(1)–GARCH(1,1) model and Gaussian innovations, a one-day-ahead 99% Value-at-Risk for losses $L_t = -R_t$ is

$$\text{VaR}_{0.99}^{\text{GARCH}}(t) = -\left(\hat{\mu}_t + \hat{\sigma}_t z_{0.01}\right),$$

where $z_{0.01}$ is the 1% quantile of the standard Normal distribution.

EVT (POT) for tail risk. To model extreme losses, we use a Peaks-Over-Threshold approach. Let u denote a high threshold (empirical 95% quantile of losses). Excesses $L_t - u$ above u are modelled by a Generalised Pareto Distribution (GPD) with scale σ_u and shape ξ. This yields closed-form expressions for high-level VaR and ES (here at 99%).

Backtesting. We evaluate risk calibration by comparing realised losses L_t with predicted VaR. We report the number of exceedances (breaches) and the empirical exceedance frequency. (Optionally) we also report the Kupiec unconditional coverage test for whether the breach frequency matches the nominal 1% level.

2.4 Functional data analysis (FPCA + AR(1) on scores)

We model each trading day as a functional observation built from high-frequency intraday prices. For day n, intraday log-prices are recorded at irregular times and interpolated using the *previous-tick* method, then mapped onto a regular grid of $J = 100$ points. The intraday cumulative log-return curve is defined by

$$Y_n(t_j) = \log P_n(t_j) - \log P_n(t_1), \qquad j = 1, \ldots, J.$$

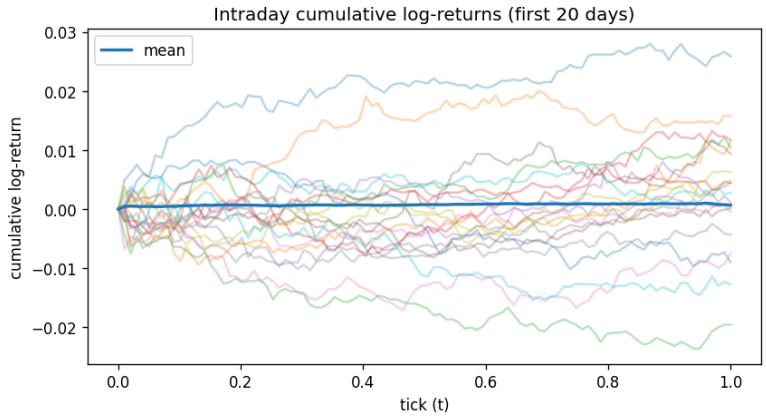

Figure 1: Intraday cumulative log-return curves $Y_n(t)$ (first 20 days) and the estimated mean function $\hat{\mu}(t)$.

We adopt the standard functional measurement-error model $Y_n(t_j) = X_n(t_j) + \varepsilon_{nj}$ with $\mathbb{E}[\varepsilon_{nj}] = 0$ and $\mathrm{Var}(\varepsilon_{nj}) = \sigma^2$, where $X_n(t)$ is a smooth latent trajectory.

The latent curve is represented via FPCA, i.e. $X_n(t) = \mu(t) + \sum_{k=1}^{p} \xi_{nk}\psi_k(t)$, with $\mu(t)$ the mean function and $\{\psi_k\}$ orthonormal eigenfunctions $\psi_k(t)$, and scores ξ_{nk}. In practice, FPCA is obtained by diagonalizing the empirical covariance of the centered discretized curves; the number of components p is chosen using the cumulative proportion of variance explained (PVE). For forecasting, we retain p_{model} as the smallest p such that PVE exceeds 90% (here $p_{\text{model}} = 2$).

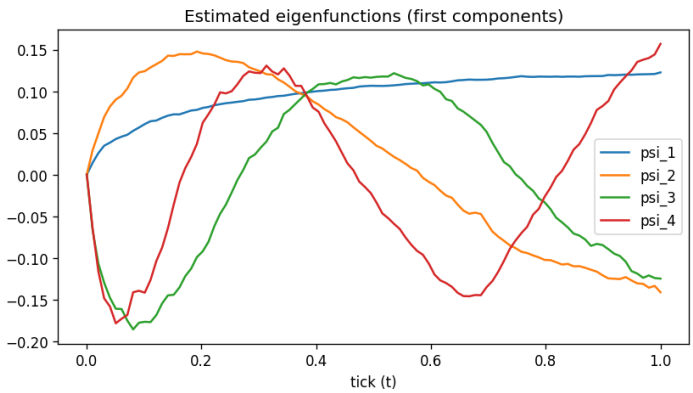

Figure 2: Estimated eigenfunctions $\hat{\psi}_k(t)$ for the first FPCA components.

To test day-ahead predictability, we fit an AR(1) model separately to each retained score sequence,

$$\xi_{nk} = \alpha_k \xi_{n-1,k} + \eta_{nk},$$

estimated on the first 80% of days and evaluated on the last 20%. One-step-ahead forecasts are $\tilde{\xi}_{n+1,k} = \hat{\alpha}_k \xi_{n,k}$, yielding the functional forecast

$$\tilde{X}_{n+1|n}(t) = \hat{\mu}(t) + \sum_{k=1}^{p_{\text{model}}} \tilde{\xi}_{n+1,k}\hat{\psi}_k(t),$$

and performance is assessed by the out-of-sample mean squared error between observed curves $Y_n(t_j)$ and forecasts $\tilde{X}_{n|n-1}(t_j)$ over the test period.

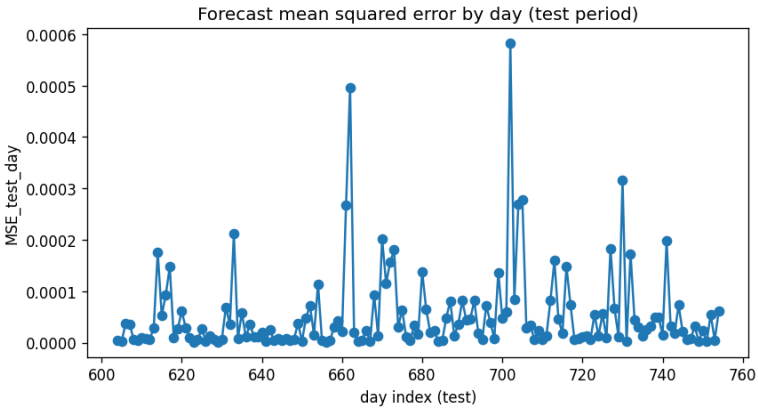

Figure 3: Out-of-sample forecast MSE per test day for the FPCA+AR(1) model.

3 Empirical Results

3.1 COVID and global volatility

Topic 1 (COVID-19 impact on US vs China)

Market	Period	Omega	Alpha	Beta	Persistence
US	Pre-COVID	7.1e-05	0.035408	0.637350	0.672758
US	Post-COVID	6.8e-05	0.033948	0.611067	0.645015
China	Pre-COVID	7.2e-05	0.035836	0.645041	0.680877
China	Post-COVID	7.1e-05	0.035426	0.637670	0.673097

Table 1: Parameters of the GARCH(1,1) model

We see that both markets show a small decrease in persistence after COVID (2020), stronger in the US and China's volatility dynamics remained stable.

Table 1 reports the estimated parameters of the GARCH(1,1) model for the United States (S&P 500) and China (CSI 300), estimated separately for the pre-COVID (2015–2019) and post-COVID (2020–2024) periods.

Across all specifications, the estimated parameters satisfy the usual non-negativity and stationarity conditions, with $\alpha + \beta < 1$.

For both markets, the parameter α is relatively small, indicating that daily volatility reacts moderately to new shocks. The GARCH parameter β is larger, suggesting that volatility dynamics are primarily driven by persistence rather than by immediate shock responses.

The persistence measure $\alpha + \beta$ decreases slightly in the post-COVID period for both countries. In the United States, persistence declines from 0.673 in the pre-COVID period to 0.645 post-COVID, while in China it decreases marginally from 0.681 to 0.673, both changes remain small. This reduction is more pronounced for the US market, indicating that volatility shocks dissipate faster after the onset of the pandemic.

Overall, these results suggest that while COVID-19 did not destabilize volatility dynamics in the long run, it coincided with a modest structural change in the speed at which volatility shocks decay, particularly in the US market.

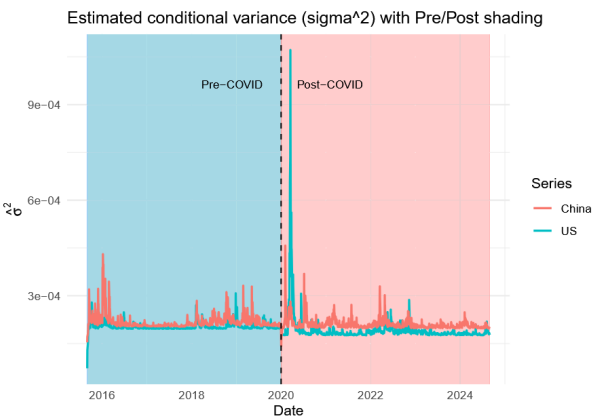

Figure 4: GARCH volatility around the COVID-19 outbreak

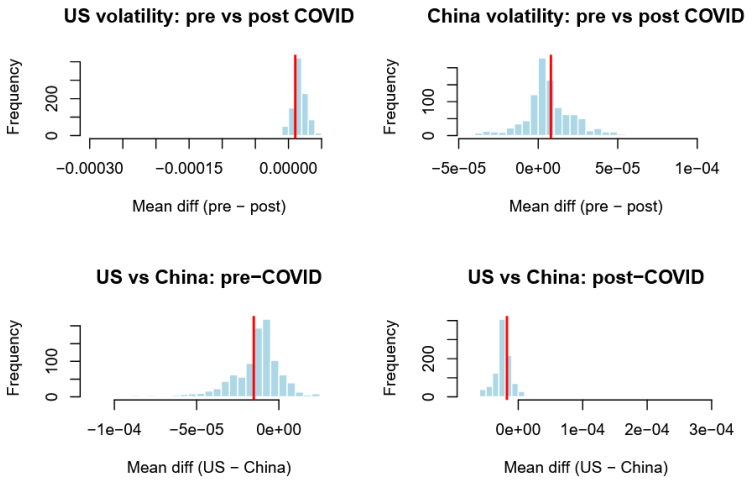

Figure 5: Histograms of mean volatility by country and period

Figure 4 illustrates the estimated conditional volatility series around the COVID-19 outbreak. A sharp and abrupt spike in volatility is observed for the US market in March 2020, corresponding to the onset of the pandemic. This episode represents an extreme but short-lived volatility shock.
In contrast, the Chinese market exhibits a more muted volatility response, with no spike of comparable magnitude.
Importantly, despite the severity of the US volatility spike, the conditional variance rapidly reverts toward its long-run level. This visual evidence is consistent with the estimated GARCH parameters.

Figure 5 presents histograms of the estimated conditional volatility for each country and period. The distributions exhibit substantial overlap, both across time and across countries. While the right tail of the US post-COVID distribution is thicker—reflecting the extreme volatility episodes during the early phase of the pandemic—the central tendency of the distributions remains similar.
These graphical results suggest that COVID-19 primarily affected the tails of the volatility distribution rather than shifting its center. In other words, the pandemic generated rare but extreme volatility events without significantly altering the long-run average level of volatility.

Test Type	Null Hypothesis (H_0)	Observed Difference	P-value
US pre vs post	$\mu_{US,pre} = \mu_{US,post}$	1.027144e-05	0.792
China pre vs post	$\mu_{China,pre} = \mu_{China,post}$	7.976756e-06	0.512
US vs China (pre)	$\mu_{US,pre} = \mu_{China,pre}$	-1.525957e-05	0.358
US vs China (post)	$\mu_{US,post} = \mu_{China,post}$	-1.755425e-05	0.755

Table 2: Block bootstrap tests on mean volatility

Formal inference on differences in mean volatility is conducted using the block bootstrap procedure. Table 2 reports the observed differences in average conditional volatility along with the corresponding bootstrap p-values.
The null hypothesis of equal mean volatility between the pre- and post-COVID periods cannot be rejected for either the US or China.
Similarly, no significant difference in mean volatility is detected between the US and China in either the pre- or post-COVID period.
These findings indicate that, despite the unprecedented market turmoil associated with the COVID-19 pandemic, there is no statistical evidence of a change in the unconditional mean volatility either over time or across the two markets.
It is important to emphasize that failure to reject the null hypothesis does not imply the absence of a COVID-19 effect altogether. Rather, it indicates that the effect did not translate into a persistent shift in the average level of volatility over the sample period.
Taken together, the results point to a clear distinction between short-run and long-run volatility effects. COVID-19 generated a severe but transitory volatility shock, particularly in the US market, as evidenced by the sharp spike in conditional variance in early 2020. However, this shock did not result in a permanent

increase in mean volatility.

3.2 Daily risk measures

3.2.1 Daily risk measures (Topic 2): exploratory analysis

Figure 6 displays the daily log returns R_t.

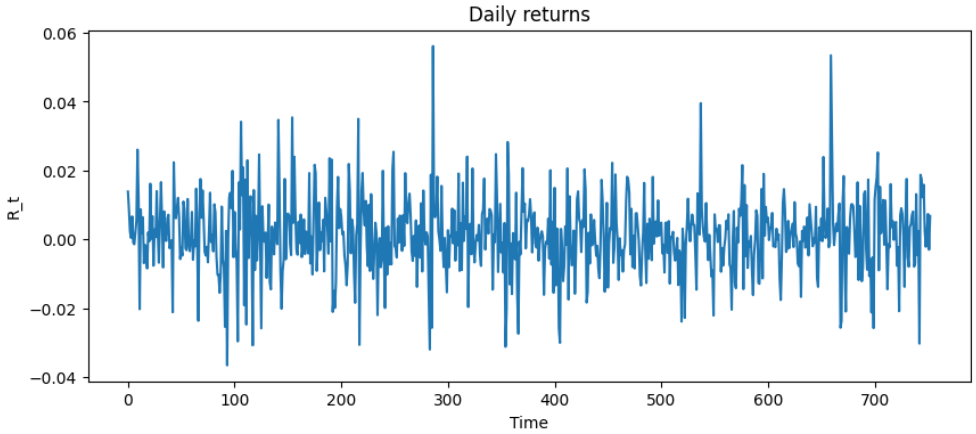

Figure 6: Daily log returns R_t.

The sample shows the usual financial stylised facts: returns are centered around zero but exhibit heavy tails. The Jarque–Bera test strongly rejects normality, consistent with excess kurtosis. Autocorrelation diagnostics confirm that while R_t itself has little serial dependence, squared returns R_t^2 exhibit significant dependence, motivating conditional heteroskedastic models such as GARCH.

3.2.2 AR(1)–GARCH(1,1): estimation and diagnostics

The estimated AR(1) coefficient is close to zero, confirming weak dependence in the conditional mean. For the conditional variance, we obtain a persistent but stationary volatility process with $\hat{\alpha} + \hat{\beta} < 1$. Diagnostics on standardised residuals show that the model removes most serial dependence in both z_t and z_t^2 (Ljung–Box p-values above conventional levels), while residual kurtosis remains above 3, indicating remaining heavy tails beyond the Gaussian assumption.

3.2.3 Volatility forecasting: GARCH versus realised variance

We compare one-step-ahead variance forecasts from the fitted GARCH model,

$$\hat{\sigma}^2_{t+1|t} = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_t^2 + \hat{\beta}\hat{\sigma}_t^2,$$

to the realised variance RV_{t+1} computed from intraday returns. Forecast errors are summarised by MSE and MAD, and we report the correlation between RV_{t+1} and $\hat{\sigma}^2_{t+1|t}$.
In our sample, the correlation is approximately **0.31**, indicating that GARCH forecasts capture part of the day-to-day variation in realised volatility, although realised variance is noisier and exhibits sharper spikes. Figure 7 illustrates that the GARCH forecast is smoother but follows the main volatility regimes.

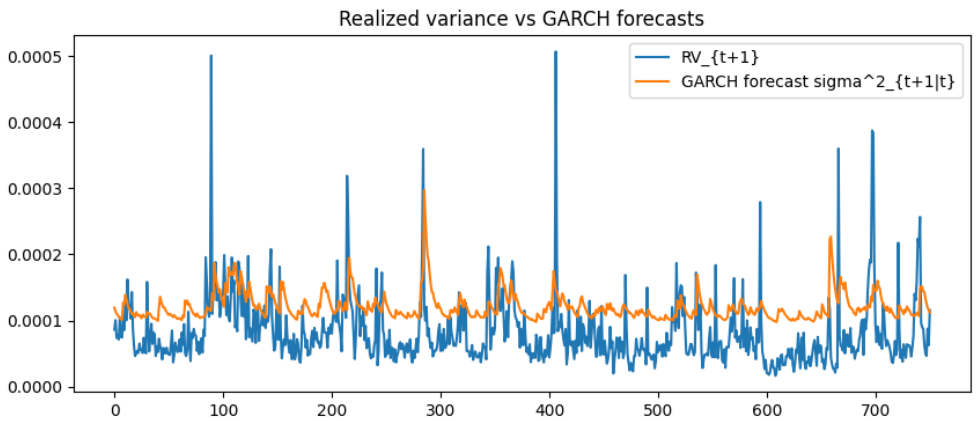

Figure 7: Realised variance RV_{t+1} and one-step-ahead GARCH variance forecasts $\hat{\sigma}^2_{t+1|t}$.

3.2.4 Extreme Value Theory (POT): tail estimation of losses

We define daily losses $L_t = -R_t$ and set the threshold u to the empirical 95th percentile (approximately $u \approx 1.75\%$), yielding 38 exceedances. This choice balances bias and variance by retaining enough tail observations while focusing on extremes. A GPD fit to excesses provides parameter estimates $\hat{\sigma}_u \approx 0.00942$ and $\hat{\xi} \approx -0.448$.

From this tail model we obtain:

- $\mathrm{VaR}_{0.99}^{\mathrm{EVT}} \approx 2.83\%$,
- $\mathrm{ES}_{0.99}^{\mathrm{EVT}} \approx 3.15\%$.

A simple in-sample backtest yields **8** exceedances over 753 days (about 1.06%), close to the nominal 1% level, suggesting reasonable calibration for extreme losses.

3.2.5 VaR comparison: GARCH–Normal vs EVT

The GARCH–Normal VaR is time-varying through $\hat{\sigma}_t$ and is typically lower than the EVT VaR, reflecting the thinner tails implied by Gaussian innovations. Over the sample, we observe **15** exceedances of the 99% GARCH–Normal VaR (about 1.99%), indicating that the Gaussian assumption tends to underestimate extreme risk.
In contrast, the EVT-based 99% VaR is more conservative and yields **8** exceedances (about 1.06%). Figure 8 overlays daily losses with both VaR measures and illustrates that breaches occur more frequently under the GARCH–Normal VaR than under the EVT VaR. Figure 8 summarises the practical implications by overlaying losses and the two competing 99% VaR estimates.

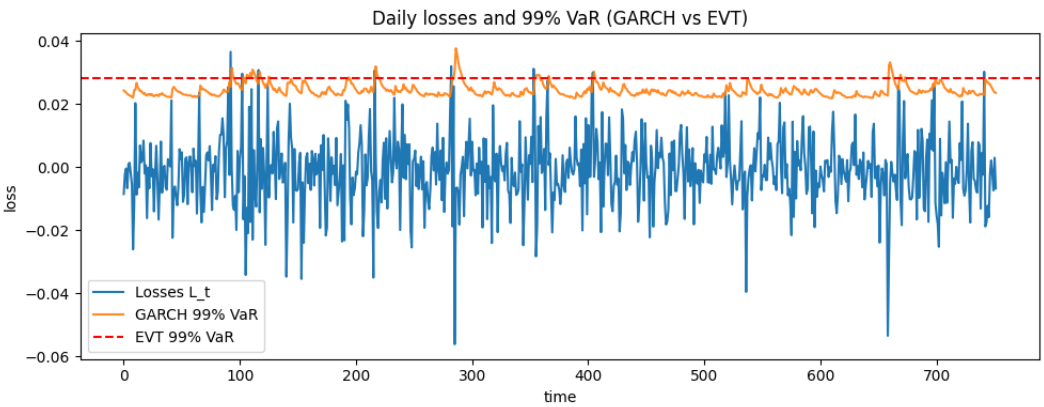

Figure 8: Daily losses L_t and 99% VaR estimates: GARCH–Normal (time-varying) versus EVT–POT/GPD (constant).

3.3 Intraday return geometry

Intraday cumulative curves. Figure 1 shows intraday cumulative log-return trajectories $Y_n(t)$ together with the mean function $\hat{\mu}(t)$. Curves fluctuate around zero, while the estimated mean curve is close to flat, suggesting that most variation arises from day-specific shapes rather than a strong deterministic intraday drift.

Eigenfunctions and explained variance. FPCA reveals a strongly low-dimensional structure: the first component explains about 87% of total variation, and the first two exceed 93% PVE. Figure 2 shows that $\psi_1(t)$ acts as a level/trend factor, while higher components capture slope and curvature (opposition between early and late trading, and possible mid-session reversals).

Score behavior and weak temporal dependence. Score distributions are close to Gaussian in the center but exhibit slightly heavier tails, consistent with financial data. Dependence diagnostics show no significant autocorrelation, and Ljung–Box tests do not reject white noise for the main components. This strongly suggests limited day-to-day predictability in intraday shape factors.

Reconstruction MSE and noise variance. The reconstruction error decreases sharply up to $p \approx 4$ and then stabilizes near 7.5×10^{-7} , which we interpret as $\hat{\sigma}^2$. Therefore, **a very small number of components (2–4) is sufficient** for a faithful representation of intraday curves.

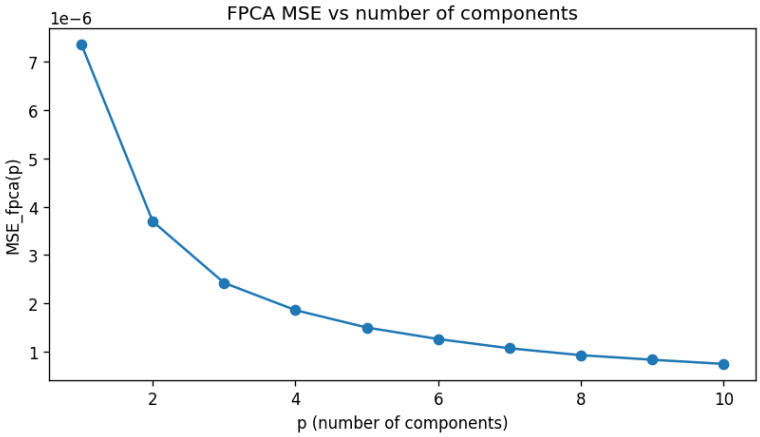

Figure 9: FPCA reconstruction MSE as a function of the number of components p.

Out-of-sample forecasting performance. Using an AR(1) model on the first two scores, the average out-of-sample forecast error is

$$\mathrm{MSE}_{\mathrm{test,mean}} \approx 5.39 \times 10^{-5},$$

corresponding to an RMSE around 7×10^{-3} (about 0.7% in cumulative log-return units). This is much larger than the in-sample reconstruction MSE ($\sim 10^{-6}$), highlighting that **intraday curve forecasting is weak**: FPCA compresses shapes well, but the temporal dynamics of scores are close to noise.

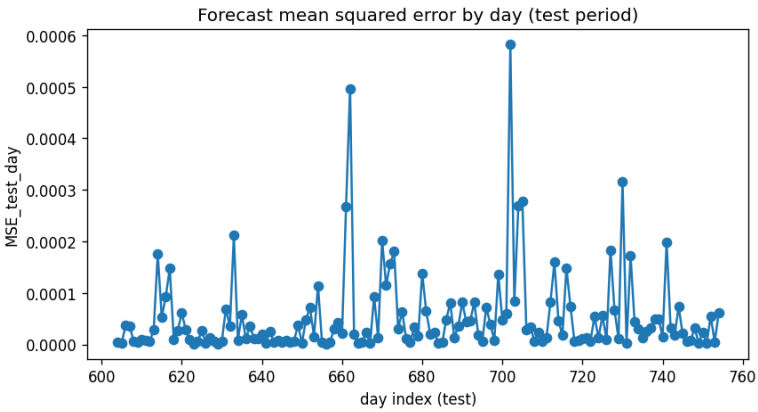

Figure 10: Out-of-sample forecast MSE per test day for the FPCA+AR(1) model.

4 Discussion and Synthesis

4.1 Market Resilience vs. Short-Run

The results for Topic 1 indicate that financial markets exhibited substantial resilience to the COVID-19 shock. While volatility increased sharply at the onset of the pandemic—particularly in the United States—this increase was short-lived and did not translate into a permanent rise in long-run volatility. The absence of a significant change in mean volatility suggests that emergency policy interventions and market adjustment mechanisms were effective in stabilizing risk conditions. At the same time, the observed reduction in volatility persistence in the US market implies a faster decay of volatility shocks. Overall, the evidence supports the view that COVID-19 constituted an extreme but temporary disruption rather than a structural break in volatility levels.

4.2 Calibration of extreme risk

Our results highlight the complementarity between conditional volatility modelling and tail modelling. The AR(1)–GARCH(1,1) specification captures volatility clustering and provides reasonable short-term

volatility forecasts when benchmarked against realised variance. However, even after standardisation, residuals remain heavy-tailed, so a Gaussian conditional distribution is too optimistic in the far tail. This is consistent with the higher-than-nominal exceedance frequency of the 99% GARCH–Normal VaR (about 2%).

The EVT-POT approach, by focusing explicitly on tail excesses, produces a more conservative VaR/ES and an exceedance frequency closer to the nominal 1%. In practice, this suggests using GARCH to track time-varying volatility while relying on EVT (or a heavy-tailed innovation distribution) to improve extreme risk calibration.

4.3 FPCA shape factors and limits of predictability

FPCA provides a compact and economically interpretable representation of intraday cumulative returns: a dominant *level/trend* factor plus a small number of *shape* factors (slope and curvature). However, despite this low-dimensional structure, the FPCA scores display very weak serial dependence: ACF and Ljung–Box diagnostics are consistent with near white-noise behavior, and AR(1) coefficients are close to zero. As a result, forecasting gains are limited and the out-of-sample error remains far above the in-sample reconstruction error. In other words, intraday curves are **well summarized** but only **weakly predictable** from past intraday shapes alone, which is consistent with the high noise-to-signal ratio typical of financial returns.

5 Conclusion

This paper combines daily and intraday data to study volatility dynamics, tail risk, and intraday return curves.

For Topic 1, COVID-19 produced a sharp but temporary volatility shock (especially in the US), without evidence of a persistent shift in mean volatility from block bootstrap tests; volatility persistence decreases slightly in the US but remains stable in China. For Topic 2, AR(1)–GARCH(1,1) forecasts track realised variance reasonably well, but Gaussian innovations underestimate extreme risk at the 99% level, while EVT-POT provides better-calibrated tail measures. For Topic 3, FPCA summarises intraday cumulative returns with very few components, but score dynamics are close to white noise, so forecasting gains from simple AR models are limited.

Overall, the results suggest that volatility models capture time-varying risk, whereas explicit tail modelling is needed for extreme losses; intraday curves are well summarised but weakly predictable. Possible extensions include Student-t innovations, GARCH–EVT combinations, or spillover models across markets.

6 Sources

- Idea for bootstrapping : *Künsch, H. R. (1989). The Jackknife and the Bootstrap for General Stationary Observations. Annals of Statistics.* It shows that standard bootstrap fails for dependent data and that a block bootstrap, where blocks of consecutive observations are resampled can be the solution.